

Adaptive Regularization and the Clay Millennium Problems: A Current Account

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Abstract

The adaptive regularization template — regularize with parameter $\alpha > 0$, bound uniformly, extract by compactness, inherit regularity — applied to the Clay Millennium Problems locates difficulty at a single step: the uniform bound. The bound step has the same character across all problems: it requires non-perturbative input that no finite-order expansion can supply. Three classes emerge. Class (A): the template closes — the Hodge Conjecture, where the Pointwise (k, k) Conjecture follows from the local $\partial\bar{\partial}$ decomposition (Hörmander for currents) and the bound step is perturbative. Class (B): the template isolates a named open conjecture — Riemann (construct the Hilbert-Pólya operator), Yang-Mills (Gribov mass $\gamma(a) \rightarrow \Lambda_{QCD}$), BSD (arithmetic Gan-Gross-Prasad for $U(r) \times U(r+1)$, $r \geq 3$), Navier-Stokes (non-concentration of vorticity, BKM). Class (C): the template fails structurally — P vs NP, where the bound step is equivalent to the problem and all three proof barriers apply.

1 The Template

1. **Regularize.** Add α -penalty; $\mathcal{P}(\alpha)$ solvable for each $\alpha > 0$.
2. **Bound.** Uniform estimate $\|u_\alpha\| \leq C$ independent of α .
3. **Compact.** Extract $u_{\alpha_n} \rightarrow u_0$.
4. **Regularity.** u_0 inherits the target property.

Steps 1, 3, 4 are typically analytic and complete. Step 2 is where all difficulty is concentrated.

2 Class (A): Hodge Conjecture [closes]

Setup. Let X be a smooth complex projective variety with Kähler form ω . A Hodge class is $\gamma \in H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X)$. BSD asks: is γ algebraic?

Template. $E_\alpha[T] = M(T) + \alpha M(T)^2$ on integral $(2k)$ -currents in class γ . Federer-Fleming gives mass-minimizers T_α for each $\alpha \geq 0$. Uniform mass bound: $M(T_\alpha) \leq M(T_{\text{ref}})$

(the α term cannot increase the mass above any competitor). Compactness: Federer-Fleming. Limit T_0 is mass-minimizing. By Almgren-De Lellis-Spadaro, singular set has codimension ≥ 2 .

Bound step (PPC). The smooth part of T_0 must be a complex k -submanifold. Proof: on a coordinate ball U , $\partial T = 0$ and $\bar{\partial} T = 0$ by bidegree. By Hörmander's $\bar{\partial}$ -Poincaré lemma for currents (extended via the closed graph theorem), $T|_U = \partial \bar{\partial} S$ for a $(k-1, k-1)$ current S . At a smooth point x of T , the current is integration over a smooth $2k$ -manifold M ; the distributional (k, k) type forces restriction of all (p, q) -forms with $(p, q) \neq (n-k, n-k)$ to M to vanish, hence $T_x M$ is a complex k -plane. Harvey-Lawson: T is calibrated by $\omega^k/k!$. Chow: closure of smooth part is algebraic. *Template closes.* ✓

3 Class (B): Named Open Conjectures

3.1 Riemann Hypothesis

Template. The Hilbert-Pólya conjecture: a self-adjoint H with $\text{Spec}(H) = \{\gamma : \zeta(\frac{1}{2} + i\gamma) = 0\}$. Self-adjointness gives real spectrum; all zeros satisfy $\sigma = \frac{1}{2}$ in one line.

Bound step. Constructing H . The GUE statistics of the zeros (Montgomery [6], Odlyzko [7]) match the eigenvalue spacing of a generic self-adjoint operator, strongly supporting existence. No construction is known. *Open: the Hilbert-Pólya operator.*

3.2 Yang-Mills Existence and Mass Gap

Template. $E_\alpha[A] = E[A] + \alpha \int |F|^4$. On finite lattice Λ , transfer matrix T_α^Λ has gap $\Delta_\alpha^\Lambda > 0$ (Perron-Frobenius on compact group). Prokhorov tightness. Gap continuous in α .

Bound step. Uniform gap $\Delta_0^\Lambda \geq \Delta_0 > 0$ (LMG). The Gribov-Zwanziger mechanism [8]: restricting to the Gribov region $\Omega = \{A : M[A] > 0\}$ gives gluon propagator $D(k^2) = k^2/(k^4 + \gamma^4)$, bounding $\Delta \geq 2\gamma^2$. The Gribov mass satisfies the self-consistent gap equation; LMG is equivalent to $\gamma(a) \rightarrow \Lambda_{QCD} > 0$ as $a \rightarrow 0$. Strong coupling: gap established by cluster expansion [9]. Weak coupling: dimensional transmutation is non-perturbative; the Balaban renormalization group [10] stalls at the non-trivial solution of the gap equation. *Open: $\gamma(a) \rightarrow \Lambda_{QCD}$; continuum Wightman limit.*

3.3 Birch-Swinnerton-Dyer

Template. $L_\alpha(E, s) = L(E, s)e^{-\alpha(s-1)^2}$; analytic rank preserved. Leibniz factorization: for K with $L(E_K, 1) \neq 0$, $L^{(r)}(E, 1) \neq 0 \iff L^{(r)}(E/K, 1) \neq 0$. Kolyvagin (rank 0) and Gross-Zagier (rank 1): unconditional [11, 12].

Bound step. For $r_{\text{an}} \geq 2$: the Generalized Gross-Zagier conjecture GGZ_r , which is the arithmetic Gan-Gross-Prasad conjecture for $U(r) \times U(r+1)$. Under arithmetic GGP, $\det(\langle \kappa^{(i)}, \kappa^{(j)} \rangle_{\text{Nek}}) \doteq L^{(r)}(E, 1)$; since this determinant is the regulator, GGZ_r implies the full BSD leading-coefficient formula. GGZ_1 : Gross-Zagier. GGZ_2 : Darmon-Rotger [13] under local hypotheses. GGZ_r , $r \geq 3$: requires p -adic L -function for $\text{GL}(2)^r$ and explicit reciprocity law on r -fold Kuga-Sato variety. Non-perturbative. *Open: arithmetic GGP for $U(r) \times U(r+1)$, $r \geq 3$.*

3.4 Navier-Stokes

Template. $\nu(\psi) = \nu_0 + \alpha|\nabla u|^2$. Smooth solutions u_α for all $\alpha > 0$ (Ladyzhenskaya $p = 4$ [14]). Testing against $-\Delta u_\alpha$ at maximum t^* of $\|\nabla u_\alpha\|^2$:

$$\nu_0 X / C_P^2 \leq C X^3 / \nu_0 + \|f\|^2 / \nu_0, \quad X = \|\nabla u_\alpha(t^*)\|^2.$$

Small data: closes. Large data: X^3 dominates.

Why X^3 and not X^2 . In dimension d , the convection term gives $CX^{d/2+1}/\nu$ at the enstrophy maximum. $d = 2$: X^2 , subcritical, closes by Gronwall (global regularity in 2D). $d = 3$: X^3 , supercritical. The exponent jump is the vortex stretching term $(\omega \cdot \nabla)u$ in the vorticity equation, absent in $d = 2$.

Bound step. By Beale-Kato-Majda [15]: blowup iff $\int_0^T \|\omega\|_{L^\infty} dt = \infty$. Tao [16]: a small modification of the NS nonlinearity (preserving energy identity and scaling) admits finite-time blowup; the actual NS nonlinearity prevents blowup by its specific algebraic structure (antisymmetry, divergence-free, Biot-Savart), but what in this structure prevents vorticity concentration is unknown. *Open: non-concentration of vorticity in L^∞ .*

4 Class (C): Structural Failure

P vs NP. For any NP-complete L , the uniform bound $\psi(L, n) = T^*(L, n)/n^k \leq C$ is equivalent to $L \in \mathbf{P}$, i.e. $\mathbf{P} = \mathbf{NP}$. No weaker bound suffices. The template relativizes, is natural (Razborov-Rudich [17]), and algebrizes (Aaronson-Wigderson [18]): no proof via the template can separate $\mathbf{P}$ from $\mathbf{NP}$. The problem is structurally distinct: not an analytic estimate but a combinatorial lower bound.

5 Classification and the Bound Step

Problem	Bound conjecture	Non-perturbative	Status
Hodge	PPC (local $\partial\bar{\partial}$)	No	Closed
Riemann	Construct H	Yes	Open
Yang-Mills	$\gamma(a) \rightarrow \Lambda_{QCD}$	Yes	Open
BSD	Arithmetic GGP, $r \geq 3$	Yes	Open
Navier-Stokes	$\int \ \omega\ _{L^\infty} < \infty$	Yes	Open
P vs NP	(Template fails)	N/A	Structural failure

Theorem 1 (Meta-theorem). *Applied to the five analytic Clay Millennium Problems, the template reduces each to a single open conjecture at the bound step, equivalent to the original problem. For four problems, the bound conjecture is non-perturbative: no finite-order expansion around any known solution produces the required estimate. For one (Hodge), the bound step is perturbative (Hörmander), and the template closes.*

Companion papers with full proofs: [19], [20], [21], [22], [23].

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